# Network Simplex II: Bounds and Starting Bases

The graph pivot from Network Simplex I assumed that arcs had no upper capacities and that a feasible tree was already available.

| Question | Today's extension |
| --- | --- |
| **An increasing arc fills up** | Include upper capacities in the ratio test |
| **The step length is zero** | Distinguish flows from the basis tree |
| **No starting tree is obvious** | Construct an auxiliary Phase I network |
| **Why do whole flows appear?** | Use integer data and incidence-matrix structure |

Complete examples come before the general rules; longer proofs and additional cases are in the appendix.

# Recall the uncapacitated pivot

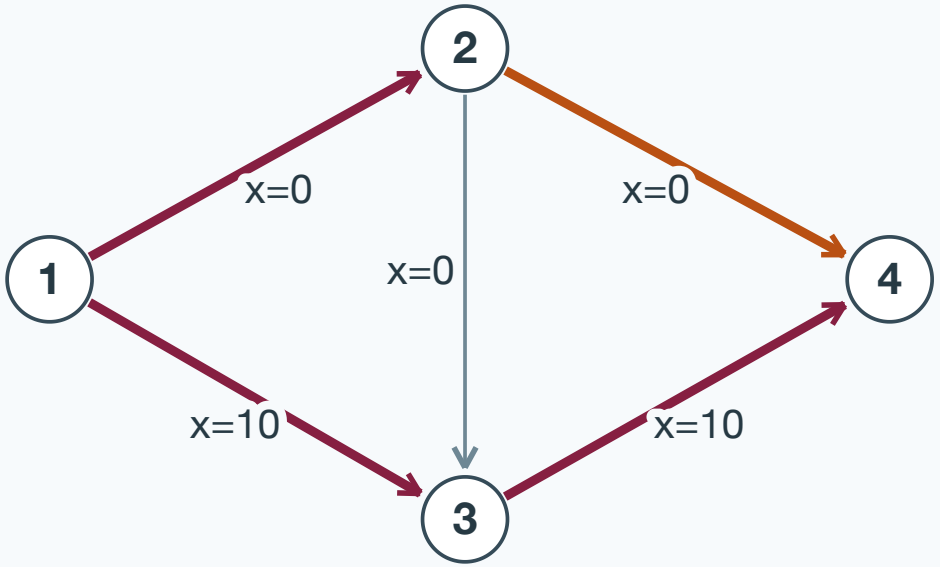
Ship 10 units from 1 to 4. Arc order $(12, 13, 23, 24, 34)$, costs $(2, 3, 2, 1, 4)$, balances $(10, 0, 0, -10)$.

Initial tree {12, 13, 34}: x = (0, 10, 0, 0, 10), cost 70.

Enter arc 24. Increasing it by θ increases $x_{12}$ and decreases $x_{13}$ and $x_{34}$.

$$x(\theta) = (\theta, 10 - \theta, 0, \theta, 10 - \theta).$$

Without upper bounds, θ = 10 and the final cost is 30. What if arc 12 can carry only 4 units?




Arrow: original arc direction. Labels: current flow. Thick maroon arcs form the basis tree.

# New data: every arc now has a capacity

Ship 10 units from 1 to 4. Arc order $(12, 13, 23, 24, 34)$, costs $(2, 3, 2, 1, 4)$, balances $(10, 0, 0, -10)$.

| Arc | 12 | 13 | 23 | 24 | 34 |
|---|---|---|---|---|---|
| Unit cost | 2 | 3 | 2 | 1 | 4 |
| Upper capacity u | 4 | 10 | 10 | 6 | 10 |
| Initial flow | 0 | 10 | 0 | 0 | 10 |

$$\min c^{\top} x \quad \text{s.t. } Ax = b, \quad 0 \le x \le u.$$

The initial tree flow is still feasible. A move can now stop when an increasing arc reaches capacity, as well as when a decreasing arc reaches zero.

# Check room to increase and room to decrease

Ship 10 units from 1 to 4. Arc order $(12, 13, 23, 24, 34)$, costs $(2, 3, 2, 1, 4)$, balances $(10, 0, 0, -10)$.

| Cycle arc | New flow | Bound on θ |
|---|---|---|
| 12 | $\theta \le 4$ | 4 |
| 24 | $\theta \le 6$ | 6 |
| 13 | $10 - \theta \ge 0$ | 10 |
| 34 | $10 - \theta \ge 0$ | 10 |

$$\theta^* = \min\{4, 6, 10, 10\} = 4.$$

Arc 12 reaches its upper capacity and leaves the basis. It becomes nonbasic at 4, not at zero. The entering arc 24 becomes basic.

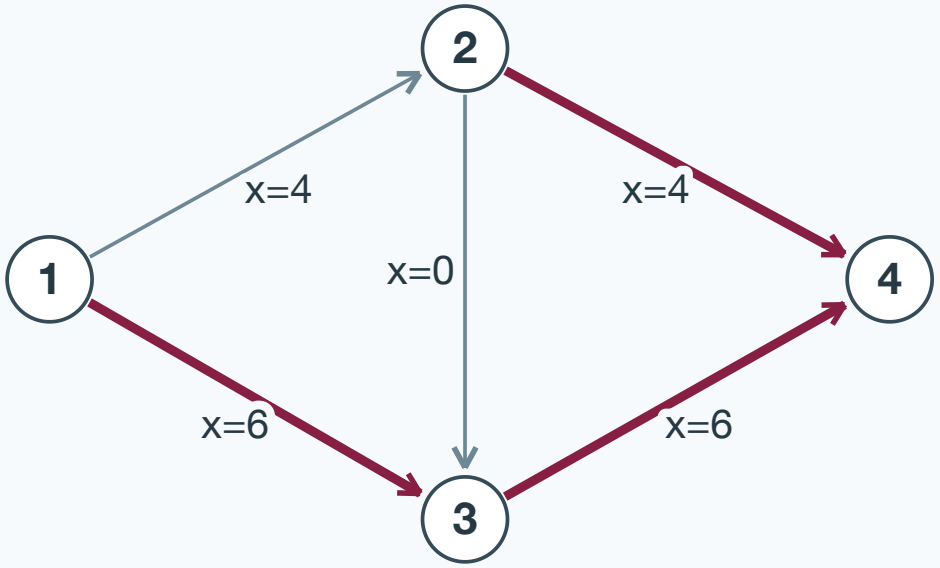
# A nonbasic variable may be fixed at its upper bound

New tree T = {13, 24, 34}. Nonbasic flows are $x_{12} = 4$ and $x_{23} = 0$.

$$x = (4, 6, 0, 4, 6), \qquad f = 54.$$

The three tree flows follow from node balances with those two nonbasic values fixed.

> Basic: solve its flow from the balances.
> Nonbasic: fix its flow at a lower or upper bound.




Arc labels: flows. Thick arcs: basic tree. Capacities in arc order (12, 13, 23, 24, 34): (4, 10, 10, 6, 10).

# The matrix form includes fixed upper-bound flows

Partition arcs into the tree T, lower-bound nonbasic arcs L, and upper-bound nonbasic arcs U. For lower bounds zero, $x_L = 0$ and $x_U = u_U$.

$$Bx_T + A_U u_U = b \quad \Rightarrow \quad x_T = B^{-1}(b - A_U u_U).$$

Here $U = \{12\}$, $u_{12} = 4$. Dropping node 4's row gives

$$b - A_{12}(4) = \begin{pmatrix} 10 \\ 0 \\ 0 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 0 \end{pmatrix}.$$

Solve the tree balances: $x_{13}$ = 6, $x_{24}$ = 4, and -$x_{13}$ + $x_{34}$ = 0 gives $x_{34}$ = 6.

# The optimality sign depends on the allowed move

Compute $p_i - p_j = c_{ij}$ on the tree and $\bar{c}_{ij} = c_{ij} - p_i + p_j$ on all arcs.

| Nonbasic flow | Allowed direction | Candidate to enter |
|---|---|---|
| **At lower bound 0** | Increase flow | $\bar{c}_{ij} < 0$ |
| **At upper bound $u_{ij}$** | Decrease flow | $\bar{c}_{ij} > 0$ |

If $\bar{c}_e \geq 0$ on L and $\bar{c}_e \leq 0$ on U, the current feasible tree flow is optimal. Tree reduced costs are zero. A chosen entering arc may still have step length zero.

An upper-bound variable with negative reduced cost is not an improving candidate: increasing it is forbidden.

# Check the bounded example's final signs

Ship 10 units from 1 to 4. Arc order $(12, 13, 23, 24, 34)$, costs $(2, 3, 2, 1, 4)$, balances $(10, 0, 0, -10)$.

Tree $\{13, 24, 34\}$ gives $p = (0, -6, -3, -7)$. Current nonbasic flows: $x_{12} = 4$ at its upper bound, $x_{23} = 0$ at its lower bound.

| Arc | Reduced cost | Allowed change / cost effect |
|---|---|---|
| **12** | $2 - 0 - 6 = -4$ | Only decrease; this increases cost |
| **23** | $2 - (-6) - 3 = 5$ | Only increase; this increases cost |

Both nonbasic signs satisfy the bounded optimality test. The flow (4, 6, 0, 4, 6) of cost 54 is optimal.

Appendix: the complete dual and the optimality identity.

# A second capacity case: the entering arc fills first

Restart from the same initial flow (0, 10, 0, 0, 10). Change the capacities to (10, 10, 10, 6, 10): only arc 12's capacity is larger than before.

$$x(\theta) = (\theta, 10 - \theta, 0, \theta, 10 - \theta), \qquad \theta^* = 6.$$

Arc 24 reaches its own upper bound before any tree arc reaches a bound.

> Keep the same tree {12, 13, 34}. Move nonbasic arc 24 from its lower-bound set to its upper-bound set. This is called a **bound flip**.

$$x = (6, 4, 0, 6, 4), \qquad f = 70 - 4(6) = 46.$$

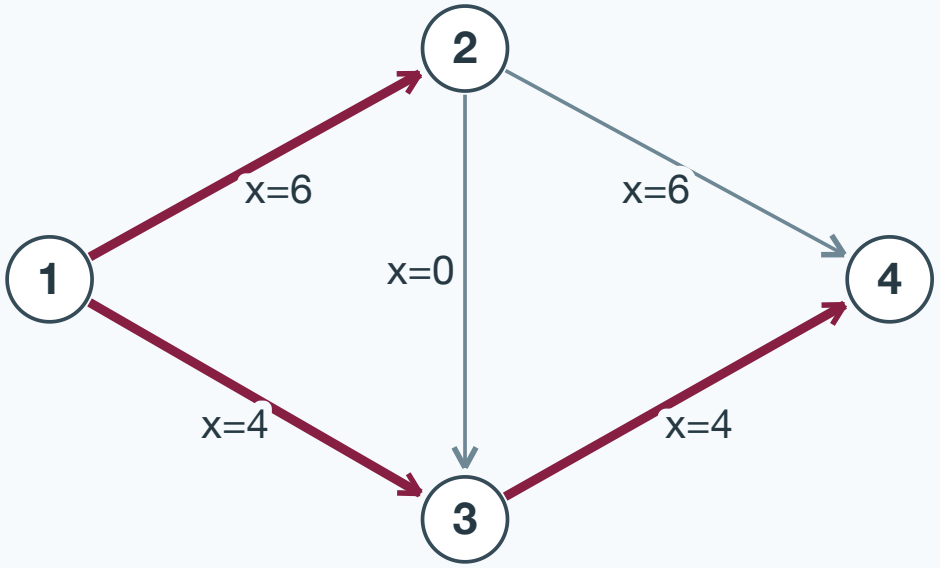
# Your turn: verify the bound-flip answer

Use costs (2, 3, 2, 1, 4), capacities (10, 10, 10, 6, 10), balances (10, 0, 0, -10).

Work for about 4 minutes.

1. Check all balances and capacities for x = (6, 4, 0, 6, 4).

2. The tree is unchanged, so p = (0, -2, -3, -7). Check the two nonbasic reduced-cost signs. Is the answer optimal?




Arc labels: flows. Thick arcs: basic tree. Capacities in arc order (12, 13, 23, 24, 34): (10, 10, 10, 6, 10).

# A bound flip can finish the problem

$$6 + 4 = 10, \quad -6 + 0 + 6 = 0, \quad -4 - 0 + 4 = 0, \quad -6 - 4 = -10.$$

Every flow is between zero and its capacity. Nonbasic arc 23 is at zero with reduced cost 1; nonbasic arc 24 is at its upper bound with reduced cost -4.

> Both signs are correct for their allowed directions. The cost 46 is optimal. A basis exchange is not required for every improving bounded-simplex step.

This completes both capacity cases. Next we return to uncapacitated flow to isolate zero-step pivots.

# The bounded cycle ratio test

Choose a lower-bound arc to increase or an upper-bound arc to decrease. Orient the cycle in that chosen change direction.

$$\theta^* = \min\left\{ \min_{e \text{ increasing}} (u_e - x_e), \quad \min_{e \text{ decreasing}} x_e \right\}.$$

Include the entering arc in this test. An infinite capacity contributes no finite restriction.

> If a tree arc reaches a bound, exchange it with the entering arc. If only the entering arc reaches its opposite bound, perform a bound flip. With tied blockers, more than one valid update may exist.

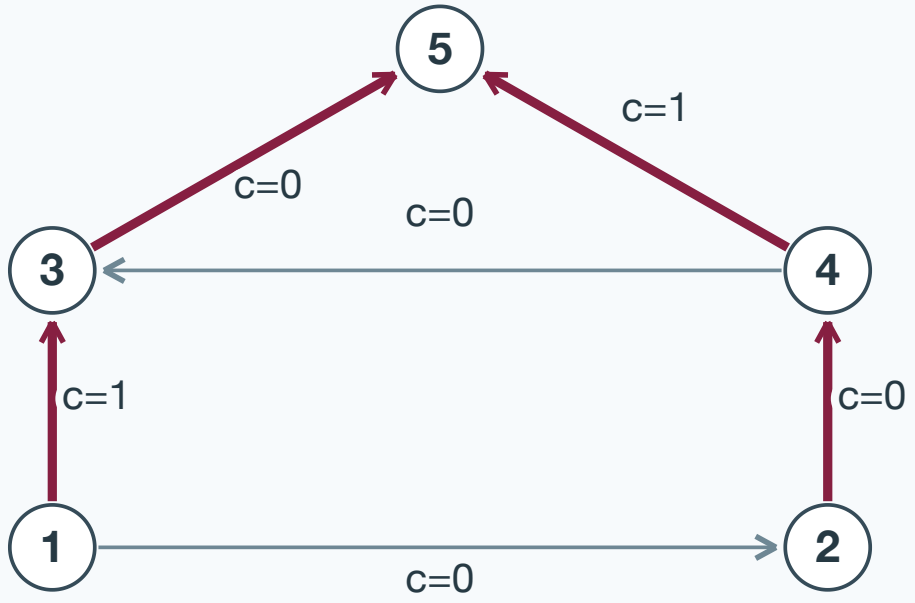
The cost change is θ times the entering reduced cost when increasing, and minus θ times that reduced cost when decreasing.

# A new example: a basis change without movement

Send one unit from node 1 to node 5. Other node balances are zero. All flows are nonnegative, with no upper bounds.

Only arcs 13 and 45 have cost 1; the other four arcs have cost 0. Thus minimize f = $x_{13} + x_{45}$.

> Initial tree {13, 24, 35, 45}. Its flows are $x_{13} = x_{35} = 1$, $x_{24} = x_{45} = 0$. The cost is 1.




Arrow: original arc direction. Labels: cost per unit. Node balances are stated beside the graph.

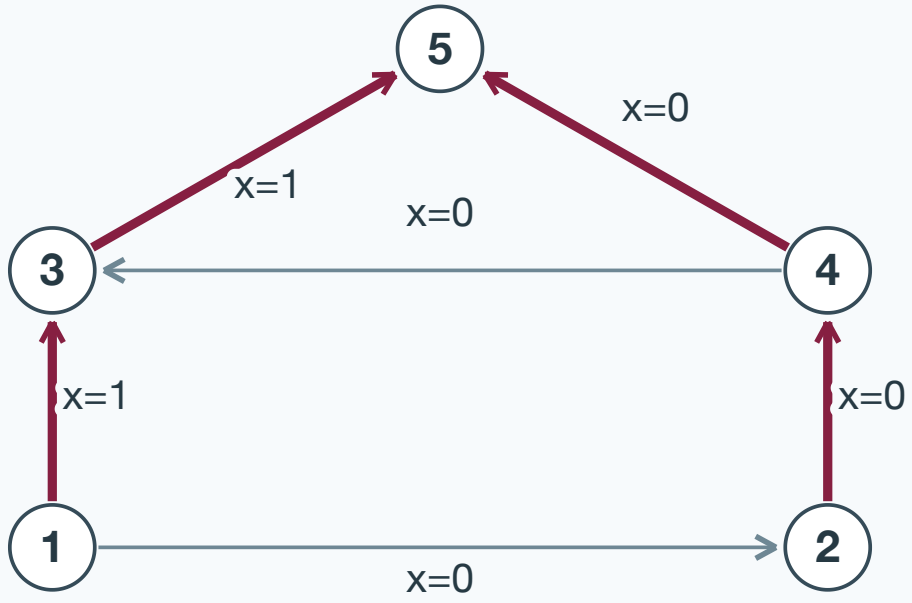
# Pivot 1: an improving direction with a zero step

Set $p_5 = 0$. The tree gives $p = (1, 1, 0, 1, 0)$, so $\bar{c}_{43} = 0 - 1 + 0 = -1$. Enter arc 43.

Cycle $4 \to 3 \to 5 \to 4$ increases $x_{43}$ and $x_{35}$, but decreases $x_{45}$.

$$x_{45} = 0 - \theta \geq 0 \quad \Rightarrow \quad \theta^* = 0.$$

> Arc 45 leaves. Flows and objective do not change, but the tree becomes {13, 24, 35, 43}.




Send one unit from 1 to 5. Labels: flows. Thick maroon arcs: the current basis tree.

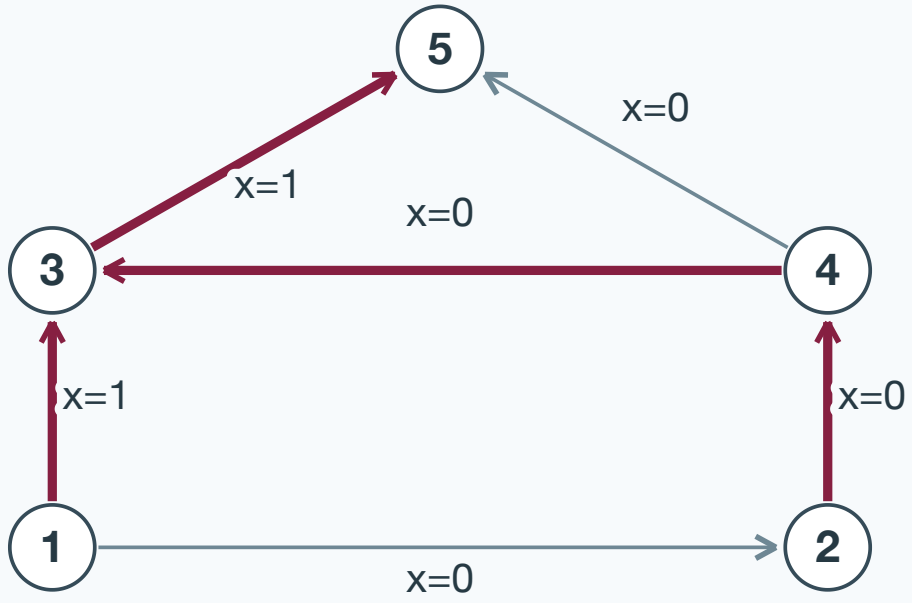
# Pivot 2: the new tree reveals a positive move

The new tree gives $p = (1, 0, 0, 0, 0)$. Now $\bar{c}_{12} = 0 - 1 + 0 = -1$. Enter arc 12.

Cycle $1 \to 2 \to 4 \to 3 \to 1$ increases $x_{12}, x_{24}, x_{43}$ and decreases $x_{13}$.

$$\theta^* = x_{13} = 1. \quad \text{Arc 13 leaves.}$$

The first zero-length pivot changed the basis information used to find this direction.




Send one unit from 1 to 5. Labels: flows. Thick maroon arcs: the current basis tree.

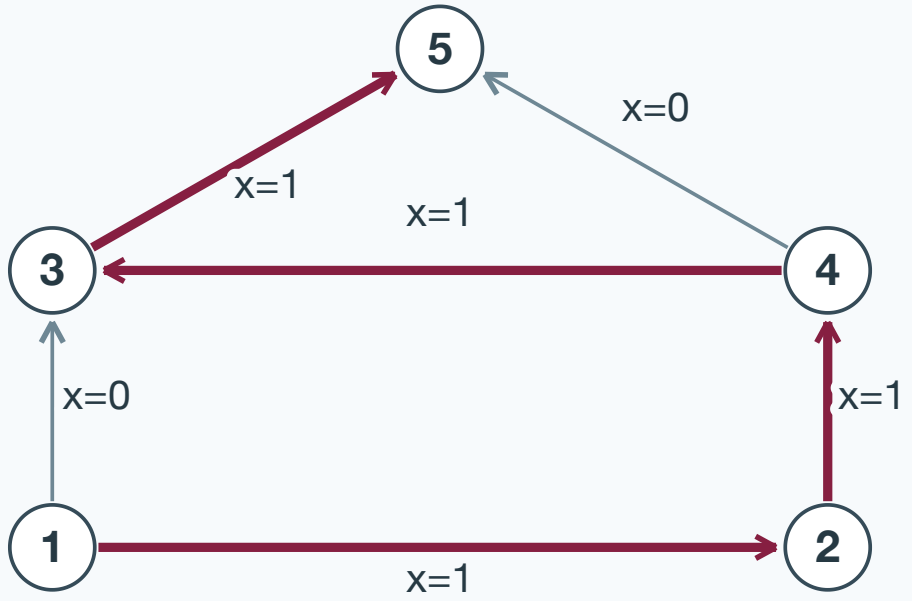
# The second pivot reaches cost zero

Final positive flows: $x_{12} = x_{24} = x_{43} = x_{35} = 1$. The unused flows $x_{13}$ and $x_{45}$ are zero.

$$f = x_{13} + x_{45} = 0.$$

All original arc costs are nonnegative, so no feasible flow can cost below zero. Equivalently, p = 0 is dual feasible and has value zero.

> A zero step does not mean that the current flow is optimal or that the algorithm should stop.




Send one unit from 1 to 5. Labels: flows. Thick maroon arcs: the current basis tree.

# Zero steps require a rule that prevents cycling

As in ordinary simplex, several bases can represent the same feasible flow. A sequence of zero-length pivots can revisit a basis.

> A proved anticycling rule is still needed for a finite-termination guarantee. Bland’s rule or a lexicographic rule can be applied to the standard-form LP.

Choosing an arbitrary tied arc is acceptable for a displayed finite calculation, but it is not by itself a general anticycling theorem.

Some network implementations use a specially rooted tree and a compatible tie rule. The next page defines the tree property; it does not replace the complete pivot rule.

# A rooted-tree property used in network implementations

Choose a node r as the root. A feasible tree is **strongly feasible toward r** if, from every other node, a sufficiently small positive amount can be sent toward r along the tree path without violating flow bounds.

| Tree arc at a bound | Required arrow direction along the path to r |
|---|---|
| **Flow 0** | Toward r, so the flow can increase |
| **Flow u > 0** | Away from r, so the flow can decrease |
| **Strictly between bounds** | Either direction works |

> This is a property of the current feasible tree and flows. It does not say that arbitrary supplies give feasible flows, or that every pivot has positive length.

Appendix: check this property in the zero-step example.

# Where does the first feasible tree come from?

A cheapest-looking spanning tree can give negative flows or violate capacities. We need a systematic start, just as in ordinary simplex.

> Use Phase I: add artificial arcs that make the balance equations easy to satisfy. Minimize total artificial flow, temporarily ignoring the original costs.

If the minimum artificial flow is zero, remove artificial arcs from the basis and restore the original objective. If it is positive, the original flow problem is infeasible.

The artificial arcs represent auxiliary variables, not real shipping routes.

# Add an artificial root and satisfy every balance

Assume $\sum_i b_i = 0$ and lower bounds are zero. Add a new root r with balance 0.

| Original node balance | Artificial arc | Initial artificial flow |
|---|---|---|
| $b_i$ > 0 | i → r | $b_i$ |
| $b_i$ < 0 | r → i | -$b_i$ |
| $b_i$ = 0 | i → r | 0 |

Set every original arc flow to zero. The artificial star is a spanning tree on the enlarged network, and all its flows are nonnegative.

$$\min w = \sum_{\text{artificial arcs } e} x_e.$$

Artificial arcs have no upper bounds. Keep the original arcs' capacity constraints; their Phase I costs are zero.

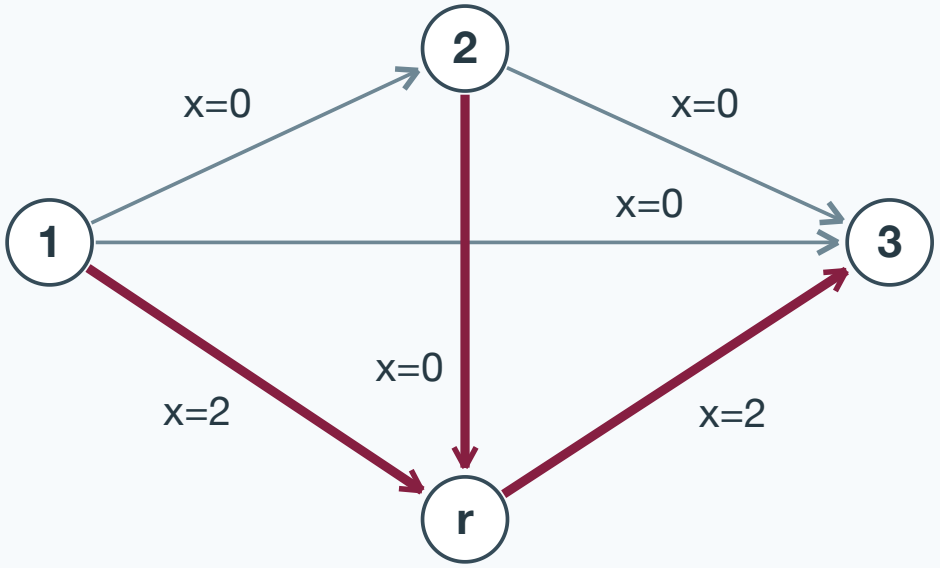
# Phase I example: two units must travel from 1 to 3

Original arcs: 12, 23, 13. Original costs: 2, 1, 5; no upper bounds. Original balances: (2, 0, -2).

Add artificial arcs 1 → r, 2 → r, r → 3. Initial artificial flows are (2, 0, 2), so w = 4.

> For Phase I, every original arc has cost 0 and every artificial arc has cost 1. The original transportation costs wait until Phase II.

The initial tree consists of the three artificial arcs. All original flows start at zero.




Auxiliary network. Arcs touching r are artificial. Labels: flows; thick arcs: current basis.

# Use the auxiliary costs to choose the first entering arc

Set the artificial root's dual variable to $p_r = 0$. The three basic artificial arcs have Phase I cost 1.

$$\begin{aligned} p_1 - p_r &= 1 \quad \Rightarrow p_1 = 1, \\ p_2 - p_r &= 1 \quad \Rightarrow p_2 = 1, \\ p_r - p_3 &= 1 \quad \Rightarrow p_3 = -1. \end{aligned}$$

| Original arc | Phase I reduced cost |
|---|---|
| 12 | $0 - 1 + 1 = 0$ |
| 23 | $0 - 1 + (-1) = -2$ |
| 13 | $0 - 1 + (-1) = -2$ |

> Choose arc 23 among the two tied improving arcs. Use auxiliary reduced costs while minimizing w.

# Phase I pivot 1: remove a zero artificial arc

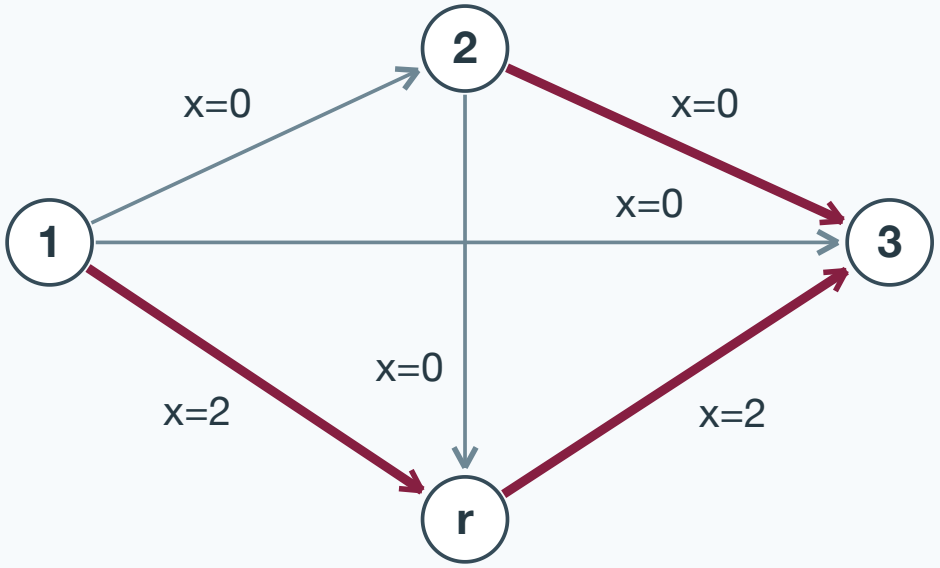
Cycle 2 → 3 → r → 2 increases $x_{23}$, decreases $x_{r3}$, and decreases $x_{2r}$.

$$\theta^* = \min\{x_{r3}, x_{2r}\} = \min\{2, 0\} = 0.$$

Arc 2r leaves; arc 23 becomes basic at zero. The new basis dual values are p = (1, -1, -1, 0).

$$\bar{c}_{12} = 0 - 1 + (-1) = -2.$$

> Artificial flow is still 4, but the new tree allows arc 12 to enter.




Auxiliary network. Arcs touching r are artificial. Labels: flows; thick arcs: current basis.

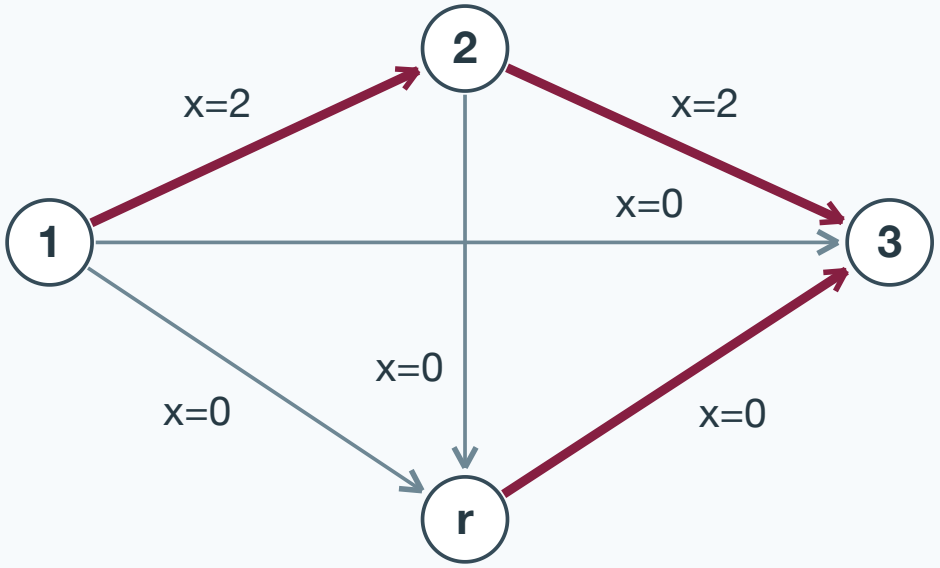
# Phase I pivot 2: eliminate the positive artificial flows

Enter 12 and follow 1 → 2 → 3 → r → 1. Increase $x_{12}$ and $x_{23}$; decrease $x_{r3}$ and $x_{1r}$.

$$\theta^* = \min\{2, 2\} = 2, \qquad w = 4 - 2(2) = 0.$$

Choose 1r to leave. The tree is now {12, 23, r3}, with flows (2, 2, 0).

> The only remaining artificial tree arc is r3 at zero. Delete it and the artificial root. The original arcs {12, 23} form a feasible tree.




Auxiliary network. Arcs touching r are artificial. Labels: flows; thick arcs: current basis.

# Restore the original costs and check optimality

Original problem: send 2 units from 1 to 3 on arcs 12, 23, 13, with costs 2, 1, 5. The feasible tree {12, 23} carries $x_{12} = x_{23} = 2$; $x_{13} = 0$.

$$f = 2(2) + 1(2) = 6.$$

Set $p_1 = 0$. Tree equations give $p_2 = -2$, $p_3 = -3$.

$$\bar{c}_{13} = 5 - 0 - 3 = 2 \geq 0, \qquad b^\top p = 2(0) - 2(-3) = 6.$$

> The original flow is already optimal. A zero Phase I objective proves feasibility; the original reduced-cost test proves optimality for the original costs.

# Your turn: why have all the flows been integers?

Work for about 4 minutes. In a tree with integer supplies and demands, start at a leaf. Its balance determines the only incident tree flow.

1. Why must that flow be an integer?
2. After removing the leaf, why is the remaining node balance still an integer?
3. With integer capacities, why is a nondegenerate cycle step also an integer?

Here “integer” means a whole number. The LP still allows real-valued flows; we are examining a property of its basic solutions.

# Integer data give integer basic feasible flows

A leaf equation fixes its one tree flow to an integer balance, possibly after a sign change. Removing the leaf updates the neighboring balance by an integer. Repeat.

For upper-bound nonbasic arcs, subtract their fixed integer flows from the node balances first. The remaining tree problem still has integer balances.

> **Integrality theorem.** With integer supplies, demands, and finite capacity bounds, every basic feasible flow is integral. If the minimum-cost flow LP has a finite attained optimum, an integral optimal flow exists.

Costs need not be integers for this existence statement. Not every optimal flow must be integral: convex combinations of different optimal whole flows can be fractional.

# The matrix reason: total unimodularity

A matrix is **totally unimodular (TU)** if every square submatrix has determinant 0, +1, or -1.

> A directed node–arc incidence matrix is TU. Deleting a redundant row, changing a column sign, or adjoining bound rows from an identity matrix preserves TU.

An invertible basis made from such a matrix has determinant ±1. Its inverse has integer entries. Multiplying by an integer right-hand side gives integer basic values.

This explains integrality for network flow, maximum flow, and assignment LPs without imposing separate integer-variable restrictions.

Appendix: determinant proof and the precise LP integrality argument.

# What you should be able to do now

| Task | Evidence to write down |
|---|---|
| **A bounded pivot** | Cycle signs, every remaining capacity, leaving arc or bound flip |
| **A zero-length pivot** | Unchanged flows; changed tree and reduced costs |
| **Phase I** | Artificial tree, objective w, and the conclusion when $w = 0$ or $w > 0$ |
| **Integral optimum** | Integer data + incidence structure, not an integer constraint |

> Network simplex is ordinary simplex using graph structure. Flow feasibility and reduced-cost optimality remain separate conditions.

Appendix: proofs, lower bounds, infeasibility and further practice.

# Appendix: complete arguments and extensions

Bounded optimality: primal and dual

A worked upper-bound entering variable

Positive lower bounds

Strong feasibility and zero steps

Phase I and removal of artificial arcs

Incidence matrices are totally unimodular

# The dual includes upper-capacity multipliers

$$\begin{aligned} \min\ & c^\top x \\ \text{s.t. } & Ax = b, \\ & 0 \le x \le u, \end{aligned}$$

$$\begin{aligned} \max\ & b^\top p - u^\top q \\ \text{s.t. } & A^\top p - q \le c, \\ & q \ge 0, \quad p \text{ unrestricted.} \end{aligned}$$

For a feasible tree satisfying the sign test, let $q_e = 0$ on T and L, and $q_e = -\bar{c}_e \ge 0$ on U. This makes the dual feasible.

$$c^\top x = p^\top b + \sum_{e \in U} \bar{c}_e u_e = p^\top b - u^\top q.$$

For the cost-54 example, p = (0, -6, -3, -7), $q_{12}$ = 4, other q = 0: the dual value is 70 - 4(4) = 54.

# The bounded sign test compares against every feasible flow

Let x be the current feasible tree flow and y any other feasible flow. Because $A(y-x)=0$,

$$c^\top (y-x) = \bar{c}^\top (y-x) = \sum_{e \in L} \bar{c}_e y_e + \sum_{e \in U} \bar{c}_e (y_e - u_e).$$

On L, both factors are nonnegative. On U, both factors are nonpositive. Every product is nonnegative; tree terms vanish because their reduced costs are zero.

Therefore $c^\top y \geq c^\top x$. This proves sufficiency of the sign test even with zero or capacity-valued basic flows.

# An upper-bound flow can enter by decreasing

Return to the capacity-4 example: x = (4, 6, 0, 4, 6), tree {13, 24, 34}, capacities (4, 10, 10, 6, 10).

Change only $c_{12}$ from 2 to 8.

Basic costs are unchanged, so p = (0, -6, -3, -7). Now the upper-bound arc 12 has reduced cost 8 - 0 - 6 = 2 > 0.

$$x(\theta) = (4-\theta,\ 6+\theta,\ 0,\ 4-\theta,\ 6+\theta).$$

All four changed arcs reach bounds at θ = 4. Choose arc 24 to leave at zero, so the tree becomes {12, 13, 34}.

$$x = (0, 10, 0, 0, 10), \quad f_{\text{old}} = 78, \quad f_{\text{new}} = 78 - 2(4) = 70.$$

# Check the changed-cost example after the pivot

Costs are now (8, 3, 2, 1, 4). The final tree {12, 13, 34} has flows (0, 10, 10); the two other arc flows are zero.

$$p = (0, -8, -3, -7), \qquad \bar{c}_{23} = 2 + 8 - 3 = 7, \quad \bar{c}_{24} = 1 + 8 - 7 = 2.$$

All nonbasic arcs are at their lower bounds with nonnegative reduced costs. The basis is degenerate because basic flows lie at bounds; the optimality proof still applies.

$$c^\top x = 70 = b^\top p.$$

> The original $c_{12} = 2$ example and this changed $c_{12} = 8$ example are different optimization problems. Recompute reduced costs whenever objective coefficients change.

# Remove positive lower bounds by a variable shift

For a problem with $\ell \le x \le u$, define $y = x - \ell$. Then

$$Ay = b - A\ell, \qquad 0 \le y \le u - \ell, \qquad c^\top x = c^\top \ell + c^\top y.$$

Thus the same zero-lower-bound algorithm applies to y, with adjusted balances and capacities. Add $\ell$ back to recover x.

> Example: arc $1 \to 2$ must carry at least 3 and at most 8. Preassign 3 units; its remaining capacity is 5. Reduce node 1's required outgoing balance by 3 and increase node 2's balance by 3.

If all bounds and balances are integers, the adjusted data are still integers. Fixed arcs with $\ell$ = u can be removed after this adjustment.

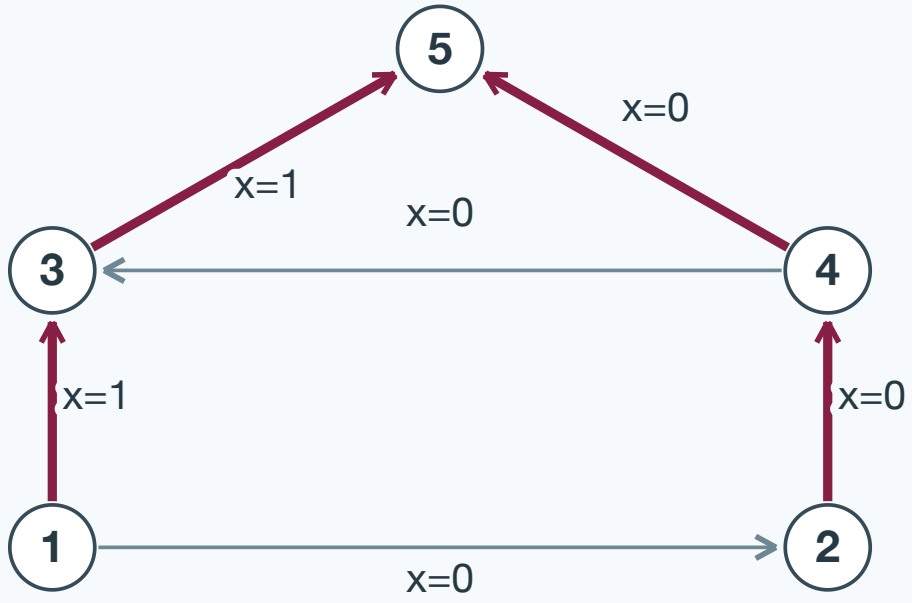
# A strongly feasible tree can still take a zero step

In the initial five-node example, choose node 5 as root. Tree arcs 24 and 45 have zero flow and point toward root 5. Positive tree flows have no upper bounds.

A small additional amount can therefore travel from each node toward 5 along the tree. The tree is strongly feasible.

> Yet entering 43 decreases zero-valued $x_{45}$ and gives θ = 0. Strong feasibility is not a statement that all improving cycle directions have positive step length.

A specialized no-cycling theorem must specify how this property is preserved when selecting a leaving arc.




Send one unit from 1 to 5. Labels: flows. Thick maroon arcs: the current basis tree.

# Why the auxiliary minimum decides feasibility

Every auxiliary variable is nonnegative, so $w \geq 0$. If the original flow problem is feasible, extend any feasible original flow by setting all artificial flows to zero. This is auxiliary-feasible with $w = 0$.

Conversely, if an auxiliary feasible flow has $w = 0$, every artificial flow is zero. Its original arc flows satisfy the original balances and capacities.

> Original feasible $\Leftrightarrow$ Phase I optimum equals zero. A positive auxiliary optimum proves infeasibility.

Because the artificial star is feasible and $w$ is bounded below, Phase I has a finite optimum. Use an anticycling rule when zero steps occur.

# Zero artificial flows may still be basic

At w = 0, remove an artificial tree arc. Its deletion divides the tree into two components. An original arc crossing that cut can replace it; the current original flow remains unchanged because this is a zero-length basis exchange.

Repeat until the original nodes are connected by original tree arcs. One remaining artificial arc attaches the artificial root at zero flow; remove that arc and the root.

> Assumption for this construction: the original graph is connected after fixed arcs are eliminated. If it has separate components, solve each component separately; every component must have balanced net supply.

This is the same zero-basic-artificial-variable issue as ordinary Phase I. Do not delete a basic column before replacing it or removing a redundant equation.

# A small infeasible Phase I example

Two nodes require b = (3, -3), but the only original arc 1 → 2 has capacity 2. Add artificial arcs 1 → r and r → 2.

$$x_{12} + x_{1r} = 3, \qquad x_{12} + x_{r2} = 3, \qquad 0 \le x_{12} \le 2.$$

$$w = x_{1r} + x_{r2} = 6 - 2x_{12} \ge 2.$$

The auxiliary optimum is 2, attained at $x_{12}$ = 2 and both artificial flows equal to 1. The original shipment requirement is infeasible: only 2 of the required 3 units can pass.

# Why every incidence subdeterminant is 0 or ±1

Consider any square submatrix M of a directed incidence matrix. Prove the claim by induction on its size; a $1 \times 1$ entry is 0, 1, or -1.

If a column has no nonzero entry, $\det M = 0$. If a column has exactly one nonzero entry, expand along it: the determinant is ± the determinant of a smaller submatrix.

Otherwise every column has two nonzero entries. They must be +1 and -1, so the sum of the rows is zero and $\det M = 0$.

> These cases exhaust all possibilities. Deleting rows or columns preserves TU. Adjoining a signed unit row or column preserves TU by the same determinant expansion.

# From determinant ±1 to integer basic solutions

For zero lower bounds, write upper capacities with slack variables: Ax = b, x + s = u, x,s ≥ 0. Remove dependent balance rows.

The coefficient matrix is built from the reduced incidence matrix by adjoining unit rows for bounds and unit columns for their slacks. It is TU.

$$B^{-1} = \frac{\operatorname{adj}(B)}{\det B}.$$

An invertible basis B has det B = ±1 and integer cofactors, so $B^{-1}$ is integer. Integer right-hand sides therefore give integer basic solutions.

> The standard-form feasible set has no lines. If its optimum is finite and attained, some basic feasible solution is optimal. Hence there exists an integral optimum.

# Integral basic solutions do not exclude fractional optima

There are two routes from s to t, each of total cost 1. Send one unit, with enough capacity on both routes.

| Feasible optimum | First route | Second route | Cost |
|---|---|---|---|
| **Whole flow A** | 1 | 0 | 1 |
| **Whole flow B** | 0 | 1 | 1 |
| **Their average** | 0.5 | 0.5 | 1 |

> The fractional average is also optimal, but it is not an extreme point of this feasible segment. The theorem guarantees an integral optimal basic solution, not that every optimum is integral.